





Exam :
Subject :
Total Marks :
QP :

End Term Quiz
LLM
50.00
Large Language Models 18 Dec 25

Exam Mode

Learning Mode

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QUESTION MENU

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01:02

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Energy: A

Samsung Galaxy Z Fold8 Ultra 512 GB in Graphite F976BZKG)

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Question 1 : 6406531859498

[View Solutions \(0\)](#) Total Mark : 2.00 | Type : MCQ

A language model produces the following logits for the next token: {apple : 2.1, banana : 1.8, grape : 0.4, orange : 0.3, melon : -1.0} Before sampling, the model applies: • Temperature scaling with $T = 0.5$ • Top-K sampling with $K = 2$ Which tokens are eligible to be sampled?

OPTIONS :

- ☐ Only "apple"
- ☐ "apple" or "banana"
- ☐ "apple", "banana" or "grape"
- ☐ "apple" or "grape"

[Check Answer](#)

[Discussions \(0\)](#)

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[Broken Format?](#)

Question 2 : 6406531859499

[View Solutions \(0\)](#) Total Mark : 2.00 | Type : MCQ



A neural network layer produces activations of hidden dimension $H = 512$. You consider two normalization strategies: 1. Batch Normalization 2. Layer Normalization Which option correctly states the number of trainable parameters added by each normalization method?

OPTIONS :

- ☐ BatchNorm adds 512 parameters, LayerNorm adds 512 parameters
- ☐ BatchNorm adds 1024 parameters, LayerNorm adds 1024 parameters
- ☐ BatchNorm adds 1024 parameters, LayerNorm adds 512 parameters
- ☐ BatchNorm adds 512 parameters, LayerNorm adds 1024 parameters
- ☐ Both add 0 trainable parameters

Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 3 : 6406531859500

View Solutions (0) Total Mark : 2.00 | Type : MCQ

A sub-word vocabulary is being constructed using the WordPiece algorithm. The following tokens and their frequencies are extracted: • replay (frequency = 100) • replus (frequency = 3) The current vocabulary includes the tokens: {**re**, **play**, **plus**, **#play**, **#plus**} Two candidate merges are under consideration: • Merge A: **re** + **play** → **replay** • Merge B: **re** + **plus** → **replus** Which of the above merges are most possible based on the WordPiece tokenizer?

OPTIONS :

- ☐ Merge **re** + **play**
- ☐ Merge **re** + **plus**
- ☐ Insufficient information to determine

Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 4 : 6406531859501

View Solutions (0) Total Mark : 2.00 | Type : MCQ

For a 4-category classification task, how would you use BERT's output?

OPTIONS :

- ☐ Use all token embeddings and average them.
- ☐ Use the [CLS] token embedding as input to a classification layer.
- ☐ Use only the [SEP] token
- ☐ Use the [MASK] token predictions.

Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 5 : 6406531859502

View Solutions (0) Total Mark : 2.00 | Type : MCQ

A hospital wants to extract disease mentions and treatment plans from unstructured clinical notes. Notes use heavy abbreviations like



A hospital wants to extract disease mentions and treatment plans from unstructured clinical notes. Notes use heavy abbreviations like "pt dx w/ CAD, rx aspirin". For extracting specific disease names from text, which task should you fine-tune BERT for?

OPTIONS :

- ☐ Masked Language Modeling
- ☐ Next Sentence Prediction
- ☐ Named Entity Recognition
- ☐ Text generation

Check Answer

Discussions (0)

Suggest Correction

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Question 6 : 6406531859503

View Solutions (0) Total Mark : 2.00 | Type : MCQ

Consider a sequence of length T and a local block attention mechanism with n blocks. If the attention matrix is restricted to only the main diagonal blocks (non-overlapping), what is the computational complexity of the self-attention operation for one layer (ignoring linear projections)? Assume d be the model dimension.

OPTIONS :

- ☐ $\mathcal{O}((T/n)^2 \cdot d)$
- ☐ $\mathcal{O}(T \cdot d^2)$
- ☐ $\mathcal{O}(T^2 \cdot d)$
- ☐ $\mathcal{O}((T^2/n) \cdot d)$

Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 7 : 6406531859504

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Match the positional encoding method (Column A) with its property/behavior (Column B). Column A (Methods) 1. APE 2. RPE 3. RoPE 4. ALiBi 5. NoPE Column B (Properties / Observations) A. Works even on sequence lengths far beyond training length without retraining, because attention bias grows proportionally to token distance. B. Even if the model sees sequences only up to length 512 during training, it fails badly at length 1024, because positions above 512 have never had embeddings learned. C. Creates an implicit rotation such that the attention score between positions i and j depends only on the difference $i - j$, not the positions themselves. D. Uses a fixed-size bank of distance embeddings such that two attention pairs (i, j) and (k, l) with the same distance use the same embedding. E. The model becomes permutation-invariant meaning it treats "cat sat mat" the same as "mat cat sat"



model becomes permutation invariant, meaning it treats cat cat cat the same as mat cat cat .

OPTIONS :

- ☐ $1 \rightarrow B, 2 \rightarrow D, 3 \rightarrow C, 4 \rightarrow A, 5 \rightarrow E$
- ☐ $1 \rightarrow D, 2 \rightarrow A, 3 \rightarrow B, 4 \rightarrow C, 5 \rightarrow E$
- ☐ $1 \rightarrow B, 2 \rightarrow C, 3 \rightarrow A, 4 \rightarrow D, 5 \rightarrow E$
- ☐ $1 \rightarrow E, 2 \rightarrow D, 3 \rightarrow A, 4 \rightarrow C, 5 \rightarrow B$

Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 8 : 6406531859505

View Solutions (0) Total Mark : 2.00 | Type : MCQ

A 2D vector $[4, 0]$ is rotated by 90° using RoPE. What is the resulting vector?

OPTIONS :

- ☐ $[0, 4]$
- ☐ $[4, 0]$
- ☐ $[0, -4]$
- ☐ $[-4, 0]$

Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 9 : 6406531859506

View Solutions (0) Total Mark : 2.00 | Type : MCQ

If a model uses RoPE, increasing the sequence length from 100 to 1000 requires:

OPTIONS :

- ☐ 900 more positional vectors
- ☐ $2 \times 1000 - 1$ RPE matrices
- ☐ No new parameters
- ☐ A new embedding matrix

Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 10 : 6406531859507

View Solutions (0) Total Mark : 2.00 | Type : MCQ

In a sequence of length $T = 6$, how many different attention pairs produce a distance of $+1$?

OPTIONS :

- ☐ 1
- ☐ 3

☐ 5☐ 6[Check Answer](#)[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)**Question 11 : 6406531859508**[View Solutions \(0\)](#) Total Mark : 2.00 | Type : MCQ

For a sequence of length $T = 6$, how many Absolute Positional Encoding (APE) vectors are needed, and how many Relative Positional Encoding (RPE) vectors are required?

OPTIONS :

☐ APE = 6, RPE = 11☐ APE = 6, RPE = 6☐ APE = 12, RPE = 11☐ APE = 5, RPE = 12[Check Answer](#)[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)**Question 12 : 6406531859509**[View Solutions \(0\)](#) Total Mark : 2.00 | Type : MCQ

A model with positional encoding crashes when the input length exceeds the training max length. Which encoding is it most likely using?

OPTIONS :

☐ ALiBi☐ APE☐ RoPE☐ RPE[Check Answer](#)[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)**Question 13 : 6406531859510**[View Solutions \(0\)](#) Total Mark : 3.00 | Type : MCQ

If the unmodified attention score is $Q_i K_j^T$, then ALiBi changes it to:

OPTIONS :

☐ $Q_i K_j^T + \text{bias}_i$ where bias is a learnable parameter.☐ $Q_i K_j^T + \beta(i + j)$ ☐ $Q_i K_j^T \cdot \gamma^{|i-j|}$ ☐ $Q_i K_j^T - \alpha|i - j|$ [Check Answer](#)



Discussions (0)

[Suggest Correction](#)[Broken Format?](#)**Question 14 : 6406531859511**

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Which of the following decoding method(s) can generate a different output every time you run the model with the same prompt?

OPTIONS :

- ☐ Greedy decoding
- ☐ Beam search with beam width = 1
- ☐ Top-p sampling with temperature $T = 1$
- ☐ Top-p sampling with temperature $T > 1$
- ☐ Beam search with beam width < 1

[Check Answer](#)

Discussions (0)

[Suggest Correction](#)[Broken Format?](#)**Question 15 : 6406531859512**

View Solutions (0) Total Mark : 2.00 | Type : MSQ

Which of the following methods modifies/modify the attention scores AFTER computing QK^T ?

OPTIONS :

- ☐ RoPE
- ☐ APE
- ☐ ALiBi
- ☐ None of these

[Check Answer](#)

Discussions (0)

[Suggest Correction](#)[Broken Format?](#)**Question 16 : 6406531859513**

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider a BART encoder with sequence length L and hidden size d_{model} (fixed). Now suppose the sequence length is doubled from L to $2L$, keeping all other parameters unchanged. Which of the following statements are true?

OPTIONS :

- ☐ Self-attention FLOPs increase by a factor of 2 .
- ☐ Self-attention FLOPs increase by a factor of 4 .
- ☐ FFN FLOPs increase by a factor of 2 .
- ☐ FFN FLOPs increase by a factor of 4 .

[Check Answer](#)

Discussions (0)

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**Question 17 : 6406531859514**

View Solutions (0) Total Mark : 2.00 | Type : SA

You are implementing a custom Block Sparse Attention kernel. Given a sequence length $T = 1024$ and block size $B = 64$, you decide to compute the main diagonal blocks plus one random off-diagonal block for each block-row. How many total $B \times B$ block computations are performed?

Answer (Numeric):

Answer

Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 18 : 6406531859515

View Solutions (0) Total Mark : 2.00 | Type : MSQ

Which of the following statements accurately describes the properties and computational complexities of the various sparse and block attention mechanisms?

OPTIONS :

- ☐ Strided Local Attention reduces the computational complexity from quadratic to linear, specifically $\mathcal{O}(cTd)$ where c is the window size.
- ☐ Dilated Attention increases the computational complexity significantly compared to standard Strided Local Attention because it covers a wider receptive field.
- ☐ Global, Random, and Local Window attention patterns handle long sequences well.
- ☐ Empirical observations suggest that random permutations of blocks are significantly more important for performance than the identity permutation.

Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 19 : 6406531859516

View Solutions (0) Total Mark : 2.00 | Type : SA

In the standard Transformer with 6 Encoder Layers, how many distinct sets of transformation weight matrices (W_Q, W_K, W_V) are learned in the entire Encoder stack, assuming 8 heads per layer?

Answer (Numeric):

Answer

Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 20 : 6406531859517

View Solutions (0) Total Mark : 2.00 | Type : SA

Suppose you are working on prefix language modeling. The sequence length is 10 and the first two tokens represent the task-specific prefix. How many non-infinity elements are there in the mask for computing attention scores?

Answer (Numeric):

Answer



Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 21 : 6406531859518

View Solutions (0) Total Mark : 2.00 | Type : SA

You are configuring a Large Language Model (LLM) for inference and need to estimate the memory footprint of the KV Cache. Calculate the KV cache memory required per token (in Bytes) given the following model architecture: • Number of Layers (L): 32 • Number of Heads (H): 16 • Head Dimension (d_{head}): 128 • Precision: 2 Bytes (FP16)

Answer (Numeric):

Answer

Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 22 : 6406531859519

Total Mark : 0.00 | Type : COMPREHENSION

A team is developing a new large language model (LLM) for code generation, focusing on an autoregressive (AR) approach using the Causal Language Modeling (CLM) objective. They implement a standard Transformer decoder-only architecture. The team is training the model on a sequence of tokens $\mathbf{x} = (x_1, x_2, \dots, x_T)$, where T is the total length. The CLM objective aims to maximize the following log-likelihood:

$$\mathcal{L}(\theta) = \sum_{t=1}^{T-1} \log P(x_{t+1} | x_1, \dots, x_t; \theta)$$

During training, they strictly enforce teacher forcing, meaning the ground-truth previous tokens are always provided as input to predict the current token. Based on the above setup, answer the given subquestions.

Broken Format?

Question 23 : 6406531859520

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : MCQ

Which of the following describes the fundamental training mechanism of this model?

OPTIONS :

- ☐ The model predicts the entire sequence $\mathbf{x}$ simultaneously without considering dependencies.
- ☐ The model predicts the token x_{t+1} based only on the tokens x_1 up to x_t .
- ☐ The model predicts a masked token x_t based on both the left and right context tokens.

Check Answer

Discussions (0)

Suggest Correction

Broken Format?

Question 24 : 6406531859521

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : MCQ

During the training phase on a sequence $\mathbf{x}$ of length T , how does the teacher forcing mechanism specifically relate to the calculation of the loss function $\mathcal{L}(\theta)$?

OPTIONS :



- ☐ It means the model uses its own previous predictions $(\hat{x}_1, \dots, \hat{x}_t)$ as input to predict x_{t+1} .
- ☐ It dictates that the loss is only calculated for tokens that were incorrectly predicted.
- ☐ It ensures that for every prediction $P(x_{t+1} | \dots)$, the model is conditioned on the ground-truth sequence $(x_1, \dots, x_t)$.
- ☐ It allows the model to see future tokens $(x_{t+2}, \dots, x_T)$ when predicting x_{t+1} .

[Check Answer](#)[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)**Question 25 : 6406531859522**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : MCQ

In the inference phase (when the model is used for generation), why is the teacher forcing technique not used, and what is the primary challenge this introduces?

OPTIONS :

- ☐ Teacher forcing is used, but only with a small probability.
- ☐ It's not used because the ground-truth sequence is unknown; this introduces the exposure bias problem, where the model is unfamiliar with its own prediction errors.
- ☐ It's not used because it drastically slows down the inference speed.
- ☐ It's not used; the challenge is the need for a separate encoder-decoder architecture.

[Check Answer](#)[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)**Question 26 : 6406531859523**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : MCQ

Which component of the Transformer decoder is essential for enforcing the Causal Language Modeling constraint?

OPTIONS :

- ☐ The Feed-Forward Network layer.
- ☐ The Positional Encoding layer.
- ☐ The masked Self-Attention layer.
- ☐ The Cross-Attention layer.

[Check Answer](#)[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)